

CSE-350 Quiz-03

ADC & DAC

Set-A

Total – 20 marks

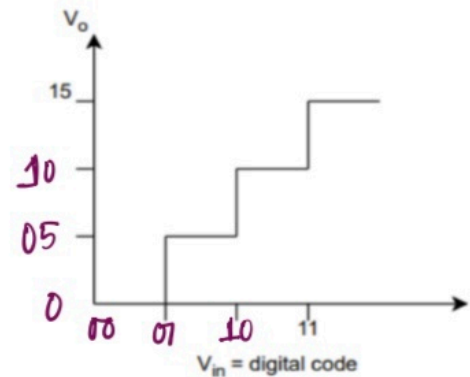
Time – 20 minutes

Name -	Id -	Section -
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Problem-1

10 marks

The figure shows the input-output characteristics of a converter. The output of the converter is denoted as V_o and the input as V_{in} . Identify the converter type and the number of bits used by the converter. Further, calculate V_{ref} when $R_F = 2R$.



Soln:

→ DAC converted
2 bits

$$R_F = 2R$$

$$\begin{matrix} b_1 b_2 \\ 1 \ 1 \end{matrix} \rightarrow V_o = 15$$

$$V_o = V_{ref} \times \frac{R_F}{R} \left(b_1 + \frac{b_2}{2} \right)$$

$$\Rightarrow 15 = V_{ref} \times \frac{2R}{R} \left(1 + \frac{1}{2} \right)$$

$$\Rightarrow 15 = V_{ref} \times 2 \times \frac{3}{2}$$

$$\therefore \boxed{V_{ref} = 5V} \quad (\text{Ans.})$$

Problem-2

10 marks

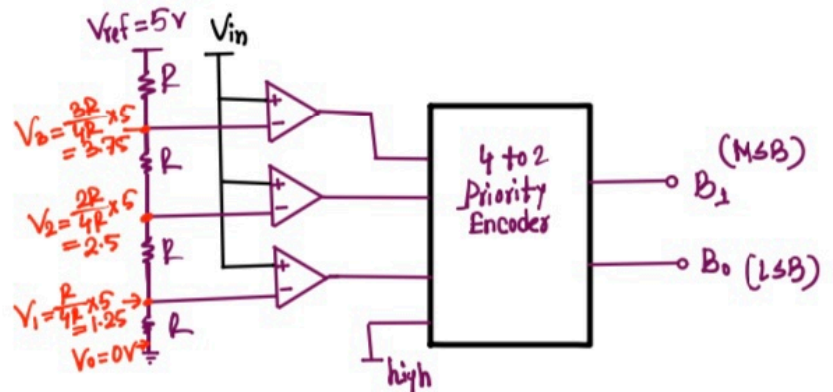
Design a 2 bit Flash ADC for an analog signal of range (0, 5). According to your circuit answer the following questions.

- What is the resolution and quantization error for this quantizer.
- Write the encoded sequences for the following input sequences { 2.3, 0.6, 4.8, 3.7 }

Soln: $n=2 \rightarrow \Delta = \frac{V_{\max} - V_{\min}}{2^n} = \frac{5-0}{2^2} = \frac{5}{4} = 1.25$

quantization error = $\frac{\Delta}{2} = 0.625$

Quantization Range	Encoded Seq.
3.75 - 5	11
2.5 - 3.75	10
1.25 - 2.5	01
0 - 1.25	00



i/p $\rightarrow \{ 2.3, 0.6, 4.8, 3.7 \}$

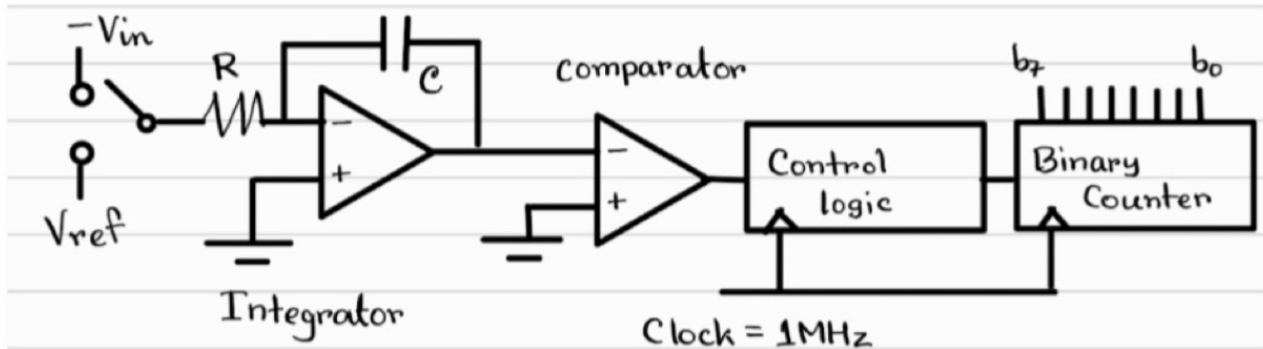
$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$

encoded $\rightarrow \{ 01, 00, 11, 10 \}$

Problem-3

05 marks

A dual slope ADC has $V_{ref} = 5V$, and a 8 bit counter that outputs an 8 bit representation of the input signal. A 1 MHz clock is used for the clock and counter circuits. $output = 01100100$



(a)	Determine the input voltage range, number of steps, and step size of the ADC.
(b)	Determine the total reading time or conversion time. 0.1100100

(a) input voltage range : $0 - V_{ref} = 5V$, $N = 8$

number of steps $= 2^N = 2^8 = 256$

Step size $= \frac{5}{256} = 0.0195V$

~~0.1100100~~
 $n = \text{Decimal of } 01100100$
 $n = 100$

(b) $f = 1MHz = 10^6 Hz$, clock cycle , $T = \frac{1}{f} = \frac{1}{10^6}$

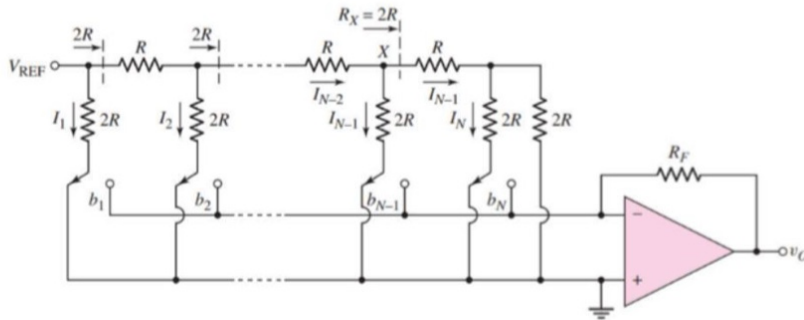
$T_1 = 2^N T = \frac{256}{10^6} = 2.56 \times 10^{-4} s$

$T_2 = n T = \frac{100}{10^6} = 1.00 \times 10^{-4} s$

$\therefore \text{Conversion time} = T_1 + T_2 = 3.57 \times 10^{-4} s$

Problem-4

05 marks



The N-bit DAC with an R-2R ladder network in the above figure is to be designed as a 3-bit DAC device. Suppose $V_{REF} = -5V$ and $R_F = R = 5K\Omega$.

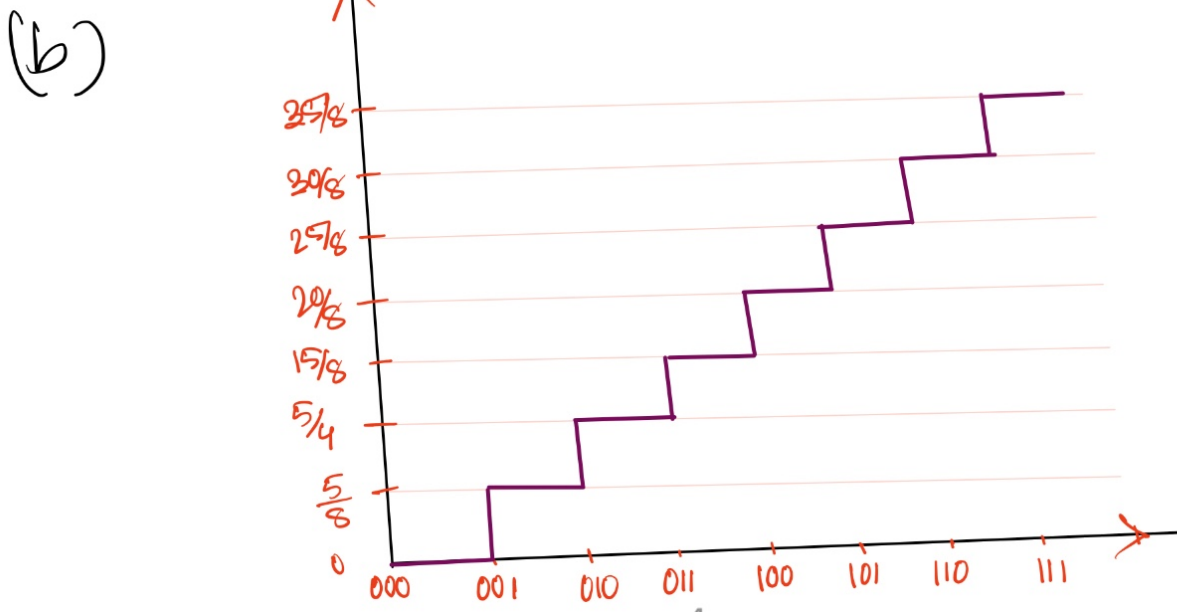
(a)	Determine the resolution and the currents I_1 , I_2 , and I_3 .
(b)	Draw the V_{in} Vs. V_{out} graph.

(a) resolution, $\Delta = -V_{ref} \frac{R_F}{2R} (b_1 + \frac{b_2}{2} + \frac{b_3}{4}) = \frac{5}{2} (0 + \frac{0}{2} + \frac{1}{4})$
 $001 \rightarrow V_O \rightarrow \therefore \Delta = \frac{5}{8}$

$$I_1 = \frac{V_{ref}}{2R} = \frac{5}{10} = 0.5mA$$

$$I_2 = \frac{I_1}{2} = 0.25mA$$

$$I_3 = \frac{I_2}{2} = 0.125mA$$



CSE-350 Quiz-03

ADC & DAC

Set-B

Total – 20 marks

Time – 20 minutes

Name -

Id -

Section -

Problem-1

10 marks

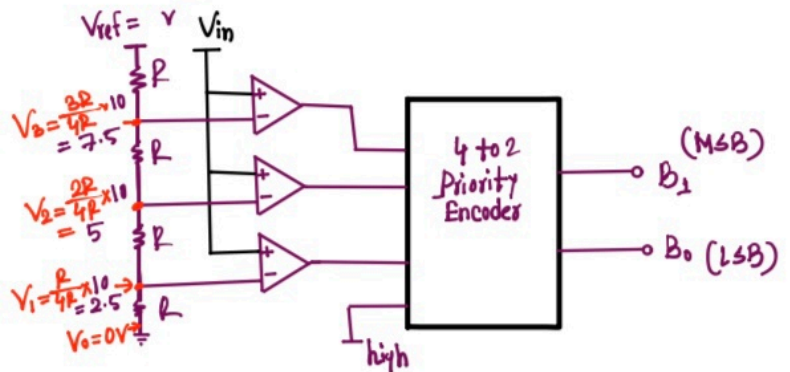
Design a 2 bit Flash ADC for an analog signal of range (0, 10). According to your circuit answer the following questions.

- What is the resolution and quantization error for this quantizer.
- Write the encoded sequences for the following input sequences { 2.3, 7.6, 5.8, 0.7 }

Soln: $n=2 \rightarrow \Delta = \frac{V_{\max} - V_{\min}}{2^n} = \frac{10 - 0}{2^2} = \frac{10}{4} = 2.5$

quantization error = $\frac{\Delta}{2} = 1.25$

Quantization Range	Encoded Seq.
7.5 - 10	11
5 - 7.5	10
2.5 - 5	01
0 - 2.5	00



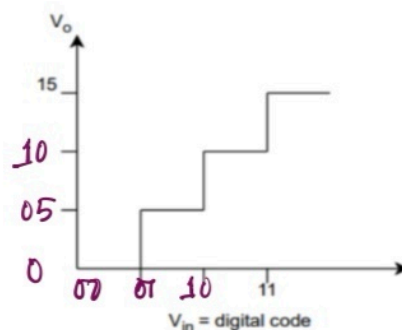
i/p $\rightarrow \{ 2.3, 7.6, 5.8, 0.7 \}$

encoded $\rightarrow \{ 00, 11, 10, 00 \}$

Problem-2

10 marks

The figure shows the input-output characteristics of a converter. The output of the converter is denoted as V_o and the input as V_{in} . Identify the converter type and the number of bits used by the converter. Further, calculate V_{ref} when $R_F = R$.



Soln: $\rightarrow$ DAC converter
2 bits

$$R_F = R$$

$$\begin{matrix} b_1 b_2 \\ 1 \ 1 \end{matrix} \rightarrow V_o = 15$$

$$V_o = V_{ref} \times \frac{R_F}{R} \left(b_1 + \frac{b_2}{2} \right)$$

$$\Rightarrow 15 = V_{ref} \times \frac{R}{R} \left(1 + \frac{1}{2} \right)$$

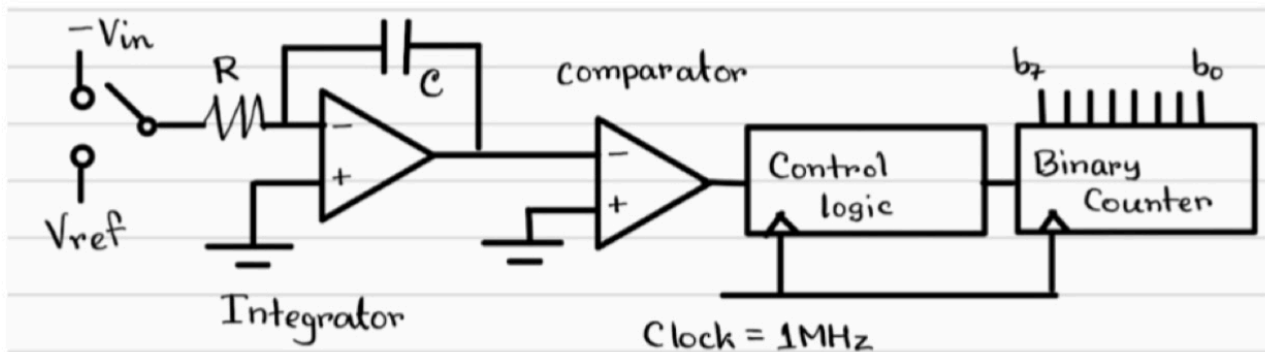
$$\Rightarrow 15 = V_{ref} \times \frac{3}{2}$$

$$\therefore \boxed{V_{ref} = 10 \text{ V}} \quad (\text{Ans.})$$

Problem-3

05 marks

A dual slope ADC has $V_{ref} = 10V$, and a 8 bit counter that outputs an 8 bit representation of the input signal. A 1 MHz clock is used for the clock and counter circuits. o/p is 01100100



(a)	Determine the input voltage range, number of steps, and step size of the ADC.
(b)	Determine the total reading time or conversion time.

(a) same as set A Q3

→ 10V

→ steps = $2^8 = 256$

→ step size = $\frac{10}{256} = 0.0390V$

(b) Conversion time = $T_1 + T_2$

$$= 2^N T + nT$$

$$= \frac{256}{106} + \frac{100}{106}$$

$$= 3.57 \times 10^{-4} s$$

$$T = \frac{1}{F} = \frac{1}{106}$$

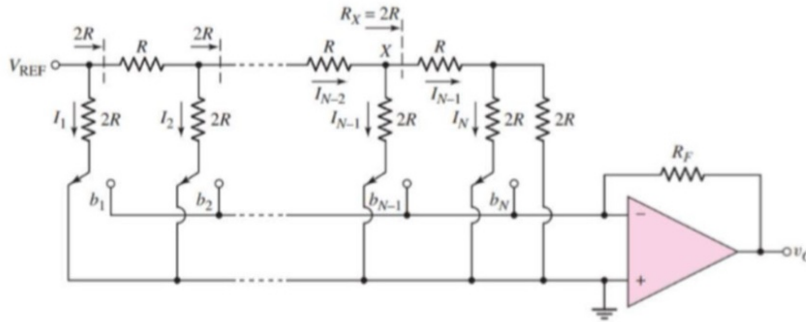
$$N=8, \text{ [redacted]}$$

$n = \text{Decimal of } 01100100$

$$n = 100$$

Problem-4

05 marks



The N-bit DAC with an R-2R ladder network in the above figure is to be designed as a 3-bit DAC device. Suppose $V_{REF} = -10V$ and $R_F = R = 10K\Omega$.

(a)	Determine the resolution and the currents I_1 , I_2 , and I_3 .
(b)	Draw the V_{in} Vs. V_{out} graph.

(a) resolution, $\Delta = -V_{ref} \frac{R_F}{2R} (b_1 + \frac{b_2}{2} + \frac{b_3}{4}) = \frac{10}{2} (0 + \frac{0}{2} + \frac{1}{4})$
 $001 \rightarrow V_o \rightarrow \therefore \Delta = \frac{10}{8}$

$$I_1 = \frac{|V_{ref}|}{2R} = \frac{10}{20} = 0.5 \text{ mA}$$

$$I_2 = \frac{I_1}{2} = 0.25 \text{ mA}$$

$$I_3 = \frac{I_2}{2} = 0.125 \text{ mA}$$

(b)

